

Oral Exam

Nicholas Eisenberg

March 29, 2022

Projects

I will briefly discuss the three following projects:

- ▶ Invariant measures for the Stochastic Heat Equation with rough initial conditions,
- ▶ Exact moment asymptotics and Solvability for the Interpolated Stochastic Heat and Wave Equation,
- ▶ Global solutions for the Interpolated Stochastic Heat and Wave Equation with a super linear diffusion term and no drift.

Project 1
Invariant Measures for the Stochastic Heat Equation

Invariant Measures for the Stochastic Heat Equation

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2} \Delta_x u(t, x) = b(x, u(t, x)) \dot{W}(t, x) \\ u(0, \cdot) = \mu(\cdot) \end{cases}$$

We assume that:

- ▶ $|b(x, u) - b(x, v)| < L_b |u - v|$ and $|b(x, 0)| \leq L_0$ for $u, v \in \mathbb{R}$ and $x \in \mathbb{R}^d$,
- ▶ W is a centered Gaussian noise that is homogeneously colored in space,

$$J(\phi, \psi) := \mathbb{E} [W(\phi) W(\psi)] = \int_0^\infty ds \int_{\mathbb{R}^d} \Gamma(dx) (\phi(s, \cdot) * \tilde{\psi}(s, \cdot))(x),$$

where $\tilde{\psi}(s, x) = \psi(s, -x)$ and Γ is a non-negative and nonnegative definite measure referred to as the *spatial correlation measure*.

Mild Solution

We interpret the solution of the Stochastic Heat Equation as the following stochastic integral equation:

$$\begin{aligned}u(t, x) &= \int_{\mathbb{R}^d} G(t, x - y) \mu(dy) + \int_0^t \int_{\mathbb{R}^d} G(t - s, x - y) b(u(s, y)) W(ds, dy) \\&= J(t, x) + I(t, x),\end{aligned}$$

where $G(t, x)$ is the heat kernel

$$G(t, x) = (2\pi t)^{-d/2} \exp\left(-\frac{|x|^2}{2t}\right)$$

A processes, $\{u(t, x) : t \geq 0, x \in \mathbb{R}^d\}$, that satisfies this relation is referred to as the *mild solution*.

Existence and Uniqueness

Existence and uniqueness of a mild solution of the Stochastic Heat Equation has been proven under the following two conditions:

- ▶ the initial measure, μ , satisfies

$$\int_{\mathbb{R}^d} \exp(-a|x|^2) \mu(dx) < \infty \quad \text{for any } a > 0,$$

- ▶ the Fourier transform of the spatial correlation measure, $\hat{f} := \mathcal{F}\Gamma$ satisfies *Dalang's condition*:

$$\Upsilon(\beta) := \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{\beta + |\xi|^2} < \infty.$$

We refer to $\hat{f}$ as the spectral measure of the noise $\dot{W}$.

Admissible Weight

Since we work in the whole space $\mathbb{R}^d$, we consider weights, ρ such that

- ▶ $\rho \in L^1(\mathbb{R}^d)$ is positive, continuous and bounded,
- ▶ for any $T > 0$, there exists a $C(T) > 0$ such that

$$\sup_{x \in \mathbb{R}^d, t \in [0, T]} \frac{(G(t, \cdot) * \rho)(x)}{\rho(x)} \leq C(T).$$

We then work in the weighted Hilbert Space, $L^2_\rho(\mathbb{R}^d)$, with norm defined as

$$\|\phi(\cdot)\|_{L^2_\rho(\mathbb{R}^d)}^2 = \int_{\mathbb{R}^d} \phi(x)^2 \rho(x) dx.$$

Some examples of admissible weights include the following:

$$\frac{1}{1 + |x|^a}, \quad a > d \quad \text{and} \quad \exp(-a|x|), \quad a > 0.$$

Intermittency

We define the upper and lower Lyapunov exponents

$$\underline{m}_2 := \liminf_{t>0} \frac{1}{t} \log \left(\mathbb{E}(|u(t, x)|^2) \right), \text{ and } \overline{m}_2 := \limsup_{t>0} \frac{1}{t} \log \left(\mathbb{E}(|u(t, x)|^2) \right).$$

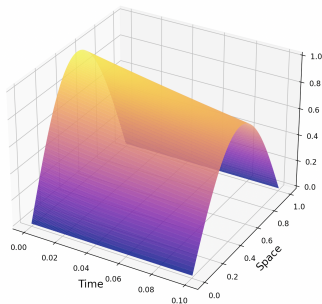
The solution is said to be *fully intermittent* when

$$\inf_{x \in \mathbb{R}^d} \underline{m}_2 > 0.$$

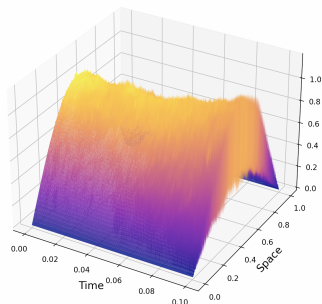
We say a *phase transition* happens when there exists $\underline{\lambda} \leq \overline{\lambda}$ such that

$$\begin{cases} \sup_{x \in \mathbb{R}^d} \overline{m}_2 = 0 & L_b < \underline{\lambda} \\ \inf_{x \in \mathbb{R}^d} \underline{m}_2 = 0 & \overline{\lambda} < l_b \end{cases}$$

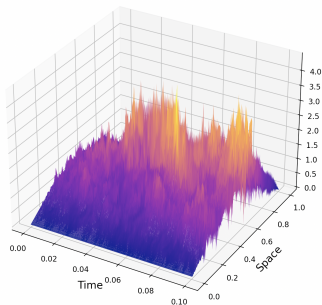
Large Peaks: $u_t - \frac{1}{2}u_{xx} = \lambda u \dot{W}$ with $u_0(x) = \sin(\pi x)$



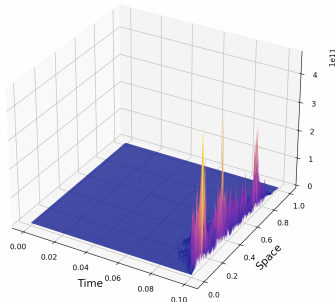
$\lambda = 0$



$\lambda = .1$



$\lambda = 1$



$\lambda = 4$

Bounded Moments

Under the assumptions for existence and uniqueness, we have the following moment bound

$$\|u(t, x)\|_p \leq C[1 + J_+(t, x)]H(t; \gamma_p)^{1/2}, \quad \gamma_p = 32pL_b^2,$$

where $t \mapsto H(t, \gamma_p)$ is nondecreasing. Following a strategy laid out by Daprato and Zabczyk, we wish to prove that

$$\sup_{t \geq 0} \mathbb{E} \left(\|u(t, x)\|_{L^2_\rho(\mathbb{R}^d)}^2 \right) < \infty.$$

Hence, a phase transition must occur. By Theorem 1.3, this forces us to require $d \geq 3$ and $\Upsilon(0) < \infty$. If we further assume that L_b is small then we may also say

$$\|u(t, x)\|_p \leq C[1 + J_+(t, x)].$$

Invariant Measures

Consider a measure space $(E, \mathcal{E})$ and denote $\mathcal{M}_1(E)$ to be the set of probability measure on this space.

Definition (Invariant Measure)

A measure $\eta(\cdot) \in \mathcal{M}_1(E)$ is invariant for transition functions $\{P_t(x, \Gamma)\}_{t \geq 0}$ if for any $\Gamma \in \mathcal{E}$ we have that

$$\eta(\Gamma) = \int_E P_t(x, \Gamma) \eta(dx) \quad \text{for any } t \geq 0$$

Recall from Calculus that the average of a function $f : [a, b] \rightarrow \mathbb{R}$ is defined as

$$\text{AVG}(f) = \frac{1}{b-a} \int_a^b f(x) dx$$

Thus, $\eta(\Gamma)$ is the average the family of probability measures $\{P_t(x, \Gamma)\}_{x \in E}$ for any $t \geq 0$.

Theorem: Existence of an Invariant measure

A processes $u(t, \cdot)(\omega)$ is bounded in probability in $L^2_\rho(\mathbb{R}^d)$ if

$$\forall \epsilon > 0, \exists R > 0 : \forall t > 0, \mathbb{P} \left\{ \|u(t, \cdot)\|_{L^2_\rho(\mathbb{R}^d)} \geq R \right\} \leq \epsilon.$$

Tessitore and Zabczyk proved the following:

Theorem

Suppose $d \geq 3$ and that there exists a $p \in [1, d/(d+2))$ such that the spatial correlation density $\gamma \in L^p(\mathbb{R}^d)$. Suppose that there exists two weights, ρ and $\hat{\rho}$ and an element $\phi \in L^2_\rho(\mathbb{R}^d) \cap L^2_{\hat{\rho}}(\mathbb{R}^d)$ such that

$$\frac{\rho}{\hat{\rho}} \in L^1(\mathbb{R}^d) \quad \text{and} \quad u^\phi(t, \cdot) \text{ is bounded in probability in } L^2_{\hat{\rho}}.$$

Then there exists an invariant measure in $L^2_\rho(\mathbb{R}^d)$.

Sketch of the Proof of Existence

It has been shown by DaPrato and Zabczyk that $P_t(\phi, \Gamma) := \mathcal{L}(u^\phi(t, \cdot))(\Gamma)$ defines a Markov transition function on $(L_\rho^2(\mathbb{R}^d), \mathcal{L})$ where,

$$\mathcal{L}(u^\phi(t, \cdot))(\Gamma) = \mathbb{P} \left[\omega \in \Omega : u^\phi(t, \cdot)(\omega) \in \Gamma \right], \quad \Gamma \in \mathcal{L}.$$

We then prove for any $\epsilon \in (0, 1)$ and $R > 0$, the existence of a compact set, $\mathcal{K} \in \mathcal{L}$ such that

$$\mathcal{L}(u^\phi(t, \cdot))(\mathcal{K}) \geq (1 - \epsilon) \mathbb{P} \left[\omega \in \Omega : \left\| u^\phi(t - 1, \cdot)(\omega) \right\|_{L_\rho^2(\mathbb{R}^d)} < R \right], \quad t \geq 1.$$

If we assume $u^\phi(t, \cdot)$ is bounded in probability in $L_\rho^2(\mathbb{R}^d)$ then one can argue the existence of a sequence $\{T_n\}$ such that $0 < T_n \uparrow \infty$

$$\frac{1}{T_n} \int_{t_0}^{T_n+t_0} \mathcal{L}(u^\phi(t, \cdot))(\Gamma) \xrightarrow[\text{weakly}]{} \eta(\Gamma) \quad \forall \Gamma \in \mathcal{L},$$

and moreover, η is invariant in $L_\rho^2(\mathbb{R}^d)$.

Tessitore and Zabczyk's Further Assumption

If $B := \sup_{t \geq 0} \mathbb{E} \left(\|u(t, x)\|_{L^2_\rho(\mathbb{R}^d)}^2 \right) < \infty$ then by Chebyshev's inequality

$$\mathbb{P}[|u(t, \cdot)|_\rho \geq R] \leq R^{-2} \mathbb{E}|u(t, \cdot)|_\rho^2 \leq R^{-2} B < \epsilon, \quad \text{for any } R \geq \sqrt{B/\epsilon}.$$

Theorem

Suppose that $d \geq 3$ and that $L < L_b^{-2}$ where

$$\tilde{\Gamma} := \left| \widehat{\gamma^{1/2}} \right| * \left| \widehat{\gamma^{1/2}} \right| \quad \text{and} \quad L := \frac{\Gamma(d/2 - 1)}{4\pi^{d/2}} \int_{\mathbb{R}^d} \tilde{\Gamma}(\zeta) |\zeta|^{2-d} d\zeta < \infty.$$

Then for any admissible weight ρ , we have that

$$\sup_{t \geq 0} \mathbb{E} \left(\left\| u^1(t, x) \right\|_{L^2_\rho(\mathbb{R}^d)}^2 \right) < \infty.$$

Moment Bound: Measure Valued Initial Conditions

Theorem

Suppose that the initial measure, μ , and spectral measure, $\hat{f}$:

$$\sup_{x \in \mathbb{R}^d} (G(t, \cdot) * |\mu|)(x) < \infty \quad \forall t > 0 \quad \text{and} \quad \Upsilon(0) < \infty. \quad (0.1)$$

Then if $\mu(dx) = \phi(x)dx$ with $\phi \in L^2_\rho(\mathbb{R}^d)$,

$$\sup_{t \geq 0} \mathbb{E} \left(\left\| u^\phi(t, x) \right\|_{L^2_\rho(\mathbb{R}^d)}^2 \right) < \infty.$$

Further, if μ has no density but we strengthen (0.1) to include

$$\int_0^s dr \, r^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-r|\xi|^2) < \infty$$

then for any $t_0 > 0$,

$$\sup_{t \geq t_0} \mathbb{E} \left(\left\| u^\mu(t, x) \right\|_{L^2_\rho(\mathbb{R}^d)}^2 \right) < \infty.$$

Invariant Measures: Measure Valued Initial Conditions

We wish prove the existence of an invariant measure for measure valued initial conditions such as the Dirac delta measure, $\delta_0(\cdot)$. Our plan is to consider

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2}\Delta_x u(t, x) = b(x, u(t, x))\dot{W}(t, x) \\ u(0, \cdot) = \mu(\cdot) \end{cases}$$

and then restart our system. In other words, we will consider

$$\begin{cases} \frac{\partial v}{\partial t}(t, x) - \frac{1}{2}\Delta_x v(t, x) = b(x, v(t, x))\dot{W}(t, x) \\ v(0, \cdot) = u(t_0, \cdot) \end{cases}.$$

With the absence of noise, we know that $v(t, x) = u(t + t_0, x)$ and we wish to show that this property holds with the addition of noise. We also know that $\sup_{t \geq 0} \|u(t + t_0, \cdot)\|_{L^2_\rho(\mathbb{R}^d)} < \infty$. Thus we need to figure out how extend the known proof to handle random initial conditions.

Project 2:

The Interpolated Stochastic Heat and Wave Equation: Solvability and Exact
Moment Asymptotics

The Stochastic Heat and Wave Equations

$$\begin{cases} (\partial_t^b - \Delta) u(t, x) = u(t, x) \dot{W}(x) & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b = 1 \quad (\text{SHE}) \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b = 2 \quad (\text{SWE}) \end{cases}$$

- ▶ $W = \{W(\phi) : \phi \in \mathcal{D}(\mathbb{R}^d)\}$ is a centered and time-independent Gaussian noise
- ▶ The choice of initial condition is such that the solution to the homogeneous equation is constant one.
- ▶ The solution is understood in the *Skorohod* sense.

Known Results

For the SHE, i.e., $b = 1$ ($a = 2$, $\nu = 2$)
[X. Chen '17]

$$\lim_{t \rightarrow \infty} t^{-\frac{4-\alpha}{2-\alpha}} \log \mathbb{E}|u(t, x)|^p = p(p-1)^{\frac{2}{2-\alpha}} (2-\alpha) \left(\frac{2\mathcal{M}}{4-\alpha} \right)^{\frac{4-\alpha}{2-\alpha}}$$

For the SWE, i.e., $b = 2$ ($a = 2$, $\nu = 2$)
[Balan, L. Chen, and X. Chen '21]

$$\lim_{t \rightarrow \infty} t^{-\frac{4-\alpha}{3-\alpha}} \log \mathbb{E}|u(t, x)|^p = p(p-1)^{\frac{1}{3-\alpha}} \left(\frac{1}{2} \right)^{\frac{\alpha}{2(3-\alpha)}} \frac{3-\alpha}{2} \left(\frac{2\mathcal{M}^{1/2}}{4-\alpha} \right)^{\frac{4-\alpha}{3-\alpha}}$$

The Interpolated Stochastic Heat and Wave Equation

$$\begin{cases} (\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r \left[\sqrt{\theta} u(t, x) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b \in (0, 1] \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b \in (1, 2) \end{cases} \quad (\text{ISHWE})$$

- ▶ ∂_t^b is the *Caputo* fractional derivative
- ▶ $(-\Delta)^{a/2}$ is the fractional Laplacian of order $a \in (0, 2]$
- ▶ I_t^r is the Riemann-Liouville fractional integral of order $r \geq 0$

The Noise, $\dot{W}$.

The time independent noise informally satisfies

$$\mathbb{E}(\dot{W}(x)\dot{W}(y)) = \gamma(x - y).$$

The spatial correlation and spectral density, γ and φ , for the noise $\dot{W}$ can be assumed to be any of the following:

$$\gamma(x) = |x|^{-\alpha}, \quad \varphi(\xi) = C_1 |\xi|^{d-\alpha} \quad \alpha \in (0, d)$$

$$\gamma(x) = \prod_{i=1}^d |x_i|^{-\alpha_i}, \quad \varphi(\xi) = C_2 \prod_{i=1}^d |\xi_i|^{1-\alpha_i} \quad \alpha_i \in (0, 1), \quad \alpha = \sum_i \alpha_i$$

Mild Solution

Definition

For $T > 0$, a random field $u = \{u(t, x) : t \in (0, T), x \in \mathbb{R}^d\}$ is called a *mild solution* if $G(t - s, x - \cdot)u(s, \cdot)1_{\{s < t\}}$ is Skorohod integrable and the following holds almost surely:

$$u(t, x) = 1 + \sqrt{\theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t - s, x - y) u(s, y) W(\delta y) \right) ds$$

where G is defined through the *Fox-H function*.

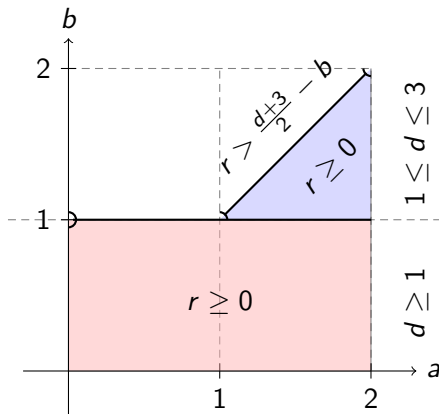
An important characteristic of G is that

$$\mathcal{FL}G(s, \xi) = \frac{s^{-r}}{s^b + \frac{\nu}{2} |\xi|^a}$$

Nonnegativity assumption on G

Under any of the following cases, G is nonnegative [Chen, Hu, Nualart '19]:

- ▶ $d \geq 1$, $b \in (0, 1]$, $a \in (0, 2]$, $r \geq 0$;
- ▶ $1 \leq d \leq 3$, $1 < b < a \leq 2$, $r > 0$;
- ▶ $1 \leq d \leq 3$, $1 < b = a < 2$, $r > \frac{d+3}{2} - b$.



Global Solution

$$(\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r \left[\sqrt{\theta} u(t, x) \dot{W}(x) \right]$$

Definition

$u(t, x)$ is a *global solution* to the (ISHWE) if $\|u(t, x)\|_p < \infty$ for any $t > 0$ and $x \in \mathbb{R}^d$.

Theorem (Chen-E. '21+)

A global solution exists provided G is nonnegative and

$$0 < \alpha < \min \left(\frac{a}{b} [2(b+r) - 1], 2a, d \right)$$

Local Solution

Definition

$u(t, x)$ is a *local solution* to the (ISHWE) if there exists $0 < T_a \leq T_b < \infty$ such that $\|u(t, x)\|_2 < \infty$ when $0 < t < T_a$ and $\|u(t, x)\|_2$ D.N.E. (does not exist) for $t > T_b$.

Theorem (Chen-E. '21+)

A local solution exists provided G is nonnegative and if

$$r \in [0, 1/2] \quad \text{and} \quad 0 < \alpha = \frac{a}{b}[2(b+r) - 1] \leq d.$$

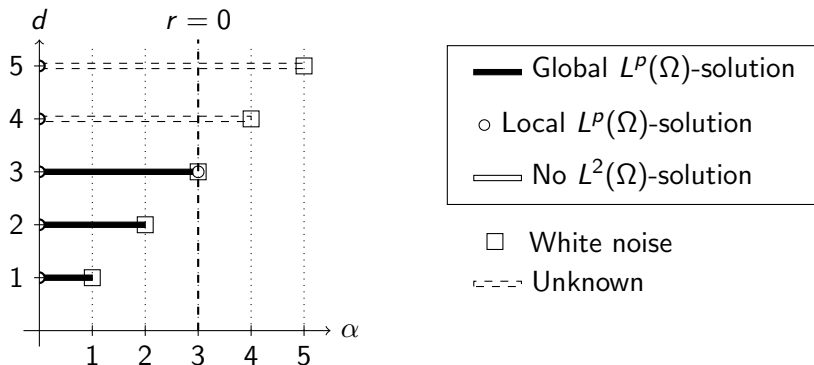
In this case, a unique $L^p(\Omega)$ solution exists for $t \in (0, T_p)$ where

$$T_p := \frac{\nu^{\alpha/a}}{2\theta(p-1)\mathcal{M}_a^{(2a-\alpha)/a}}$$

and the solution does not exist for $t > T_2$.

Example: Solvability for the SWE ($a = b = 2$)

A local solution only exists when $\alpha = 3 + 2r \leq d \leq 3$.

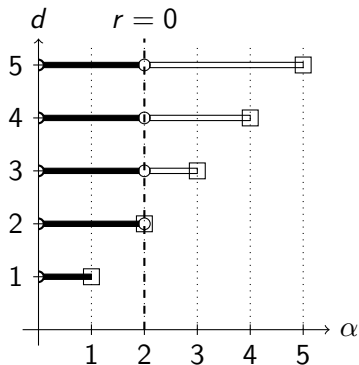


By replacing $\nu = 2$ in the following, we recover (1.12) [Balan, L. Chen & X. Chen '21]

$$T_p = \frac{\nu^{3/2}}{2\theta(p-1)\sqrt{\mathcal{M}_{2,3}(\delta_0)}}, \quad p \geq 2$$

Example: Solvability for the SHE ($a = 2$ and $b = 1$)

By setting $a = 2$, $b = 1$ and $r = 0$, we obtain the following condition for existence of a local solution: $\alpha = 2 \leq d$.



- ▶ When $\alpha = d = 2$, the critical time becomes
$$T_p = \frac{\nu}{2\theta(p-1)\mathcal{M}_{2,2}(\delta_0)}.$$
- ▶ Theorem 4.1 [Y. Hu, '01] proves that an $L^2(\Omega)$ solution exists for $t < 2$ but not for $t > 2\pi$.
- ▶ T_2 being precise implies
$$2 \leq T_2 = \frac{1}{2\mathcal{M}_{2,2}(\delta_0)} \leq 2\pi$$

Wiener Chaos Expansion

Through a standard procedure, we may define

$$f_n(x_1, \dots, x_n; x, t) = \int_0^t \int_0^{t_n} \cdots \int_0^{t_2} \prod_{k=1}^n G(t_{k+1} - t_k, x_{k+1} - x_k) dt_1 \cdots dt_n$$

where $t = t_{n+1}$ and $x = x_{n+1}$, and say that

1. $u(t, x) = 1 + \sum_{k=1}^{\infty} \theta^{k/2} I_k(f_k(\cdot, x, t)), \quad (t, x) \in (0, T) \times \mathbb{R}^d$
2. $\mathbb{E}(u(t, x)^2) = \sum_{k=0}^{\infty} \theta^k \|f_k(\cdot, x, t)\|_{\mathcal{H}^{\otimes n}}^2, \quad (t, x) \in (0, T) \times \mathbb{R}^d.$

Through a change of variable, one can easily show that

1. $\mathcal{F}G(t, \cdot)(c\xi) = c^{-\frac{a}{b}(b+r-1)} \mathcal{F}G\left(c^{\frac{a}{b}}t, \cdot\right)(\xi)$
2. $\left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = t^{[2(b+r)-b\alpha/a]n} \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2$

How to Find the Critical α

For simplicity we consider the SWE ($a = b = 2$ and $r = 0$).

$$\begin{aligned}\|u(t, x)\|_2^2 &= \sum_{n \geq 0} \theta^n n! \overbrace{t^{[4-\alpha]n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2}^{\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2} \\ &\leq \sum_{n \geq 0} \frac{(\theta t^{4-\alpha} c_1 2^{4-\alpha} C_\mu)^n}{(n!)^{3-\alpha}}, \quad C_\mu = \int_{\mathbb{R}^d} \left(\frac{1}{1 + |\xi|^2} \right)^2 \mu(d\xi) < \infty\end{aligned}$$

When $\alpha = 3$, then the above reduces down to, we lose the $n!$ term in the denominator.

How to Find the Blowup Time (SWE)

Recall from above that the critical case corresponds to $\alpha = 3 = d$.

$$\begin{aligned}\|u(t, x)\|_2^2 &= \sum_{n \geq 0} \theta^n n! \left\| \tilde{f}(\cdot, 0, t) \right\|_{H^{\otimes n}}^2 \\ &= \sum_{n \geq 0} (\theta t)^n n! \left\| \tilde{f}(\cdot, 0, 1) \right\|_{H^{\otimes n}}^2 \\ &=: \sum_{n \geq 0} (\theta t)^n R_n, \quad R_n = n! \left\| \tilde{f}(\cdot, 0, 1) \right\|_{H^{\otimes n}}^2.\end{aligned}$$

The lemma below says that $\frac{1}{n} \log R_n \rightarrow \log(2\rho)$. Now the Cauchy-Hadamard theorem can be directly applied to see that the radius of convergence is precisely $(2\theta\rho)^{-1} = T_2$.

Recall the scaling property

$$\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = t^{(4-\alpha)n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2.$$

Using this scaling property we see that

$$\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt = \Gamma((4-\alpha)n+1) \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2.$$

Lemma (R. Balan, L. Chen & X. Chen '21)

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\Gamma((4-\alpha)n+1) \left\| \tilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log [2^{4-\alpha} \rho]$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left((n!)^{4-\alpha} \left\| \tilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log \left[\left(\frac{2}{4-\alpha} \right)^{4-\alpha} \rho \right]$$

ρ

We define

$$\rho_{\nu,a}(\gamma) = \sup_{\|f\|_{L^2(\mathbb{R}^d)}=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f(x+y)f(y)}{\sqrt{1+\frac{\nu}{2}|x+y|^a} \sqrt{1+\frac{\nu}{2}|y|^a}} dy \right]^2 \mu(dx)$$

Theorem (X. Chen '07)

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] \\ = \log(\rho_{\nu,a}(\gamma)).$$

where γ is the spatial correlation function.

Connection of the SPDE with ρ

Recall from above that

$$\mathcal{FLG}(1, \xi) = \frac{1}{1 + \frac{\nu}{2} |\xi|^a}.$$

By using this relation, it can be shown that

$$\mathcal{FL}(\tilde{f}_n)(1, \xi) = \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a}.$$

Applying this to the limit representation of ρ above yields that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\int_{(\mathbb{R}^d)^n} \left| \mathcal{FL}(\tilde{f}_n)(1, \xi) \right|^2 \mu(d\vec{\xi}) \right] = \log(\rho_{\nu, a}(\gamma)).$$

$\mathcal{M}$

We define

$$\begin{aligned}\mathcal{M}_{a,d}(\gamma, \theta) &:= \sup_{g \in \mathcal{F}_a} \left\{ \left(\iint_{\mathbb{R}^{2d}} g^2(x) g^2(y) \gamma(x+y) dx dy \right)^{1/2} - \frac{\theta}{2} \mathcal{E}_a(g, g) \right\} \\ &= \sup_{g \in \mathcal{F}_a} \left\{ \langle g^2 * g^2, \gamma \rangle_{L^2(\mathbb{R}^d)}^{1/2} - \frac{\theta}{2} \mathcal{E}_a(g, g) \right\},\end{aligned}$$

where

$$\begin{aligned}\mathcal{E}_a(g, g) &:= (2\pi)^{-d} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}g(\xi)|^2 d\xi \quad \text{and} \\ \mathcal{F}_a &:= \left\{ f \in L^2(\mathbb{R}^d) : \|f\|_{L^2(\mathbb{R}^d)} = 1, \mathcal{E}_a(f, f) < \infty \right\}.\end{aligned}$$

Theorem (Bass, X. Chen and Rosen '09)

$$\rho_{\nu,a}(\gamma) = \nu^{-\alpha/a} \mathcal{M}_a^{2-(\alpha/a)}(\gamma) < \infty$$

The Moment Asymptotic

Theorem (Chen-E. '21+)

Suppose a global solution to the SPDE in question exists and $\dot{W}$ is given through the generalized Riesz kernel defined above. Then,

$$\begin{aligned} & \lim_{t_p \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p \\ &= \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^\beta \\ & \quad \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}\right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right), \end{aligned}$$

where

$$\beta := \frac{2(b+r) - \frac{b\alpha}{a}}{2(b+r) - \frac{b\alpha}{a} - 1} \quad \text{and} \quad t_p := (p-1)^{1-1/\beta} t.$$

Exact Moment Lyapunov Exponent

(Recall $\beta := \frac{2(b+r) - \frac{b\alpha}{a}}{2(b+r) - \frac{b\alpha}{a} - 1}$)

Corollary (Chen-E. '21+)

For $p \geq 2$ fixed, we have that

$$\begin{aligned} & \lim_{t \rightarrow \infty} t^{-\beta} \log \mathbb{E}(|u(t, x)|^p) \\ &= p(p-1)^{\frac{1}{2(b+r) - \frac{b\alpha}{a} - 1}} \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^{\beta} \\ & \quad \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}\right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right). \end{aligned}$$

From this we can deduce that $t \mapsto \mathbb{E}(|u(t, x)|^p)$ will grow like $\exp(K_1 t^\beta)$, in other words, the moments will grow exponentially in time, and we obtain the exact expression for the constant K_1 .

Exact Large Moment Asymptotics

(Recall $\beta := \frac{2(b+r) - \frac{b\alpha}{a}}{2(b+r) - \frac{b\alpha}{a} - 1}$)

Corollary (Chen-E. '21+)

For $t > 0$ fixed, we have that

$$\begin{aligned} & \lim_{p \rightarrow \infty} p^{-\beta} \log \mathbb{E}(|u(t, x)|^p) \\ &= t^\beta \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^\beta \\ & \quad \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}\right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right). \end{aligned}$$

From this we can deduce that $p \mapsto \mathbb{E}(|u(t, x)|^p)$ will grow like $\exp(K_2 p^\beta)$, in other words, the moments will grow exponentially in p as well.

References



Balan, R. M., Chen, L., and Chen, X. (2021).

Exact asymptotics of the stochastic wave equation with time-independent noise.

Ann. Inst. Henri Poincaré Probab. Stat. , to appear.



Bass, R., Chen, X., and Rosen, M. (2009).

Large deviations for Riesz potentials of additive processes.

Ann. Inst. Henri Poincaré Probab. Stat. **45**, 626–666.



Chen, L. and Eisenberg, N. (2021+).

Interpolating the Stochastic Heat and Wave Equations with Time-independent Noise: Solvability and Exact Asymptotics

arXiv:2108.11473, submitted.



Chen, X. (2017).

Moment asymptotics for parabolic Anderson equation with fractional time-space noise: in Skorohod regime.

Ann. Inst. Henri Poincaré: Prob. Stat. **53** 819–841.

References



Hu, Y. (2001).

Heat equation with fractional white noise potential.

Appl. Math. Optim. **20** 221–243.

Project 3: Global Solutions for the Interpolated Stochastic Heat and Wave Equation with a Superlinear Drift Term

The Interpolated Stochastic Heat and Wave Equation

$$\begin{cases} (\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r [\sigma(u(t, x)) \dot{W}(x)] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b \in (0, 1] \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b \in (1, 2) \end{cases} \quad (0.2)$$

The drift coefficient has super-linear growth:

$$|\sigma(x)| \leq \sigma_1 + \sigma_2 |x| [\ln(|x|)]^\delta \quad \text{as } |x| \rightarrow \infty.$$

We first consider the noise to be space-time white. In other words, the covariance structure is defined for $\phi, \psi \in C_0^\infty(\mathbb{R}_+ \times \mathbb{R}^d)$ as

$$J(\phi, \psi) := \mathbb{E}(W(\phi)W(\psi)) = \int_0^\infty ds \int_{\mathbb{R}^d} dx \phi(s, x)\psi(s, x).$$

Mild Solution

For any $t \in [0, T]$ and $x \in \mathbb{R}^d$, we write

$$u(t, x) = 1 + I(t, x)$$

where

$$I(t, x) = \int_0^t \int_{\mathbb{R}^d} Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy).$$

The function Y is defined through the Fox-H function and its Fourier transform can be shown to be given as

$$\mathcal{F}Y(t, \cdot)(\xi) = t^{\beta+\gamma-1} E_{\beta, \beta+\gamma}(-2^{-1} \nu t^\beta |\xi|^\alpha),$$

where

$$E_{\alpha, \beta}(z) = \sum_{k \geq 0} \frac{z^k}{\Gamma(\alpha k + \beta)}.$$

Existence and Uniqueness

Because of the bounded initial conditions, [1, Theorem 3.1] implies that the following condition on Y will imply existence and uniqueness:

$$\int_0^t ds \int_{\mathbb{R}^d} dy |Y(s, y)|^2 < \infty, \quad \text{for all } t > 0,$$

which is equivalent to

$$d < 2\alpha + \frac{\alpha}{\beta} \min\{2\gamma - 1, 0\} =: \Theta,$$

which is further equivalent to

$$\rho(d) > 0 \quad \text{and} \quad d < 2\alpha,$$

where

$$\rho(x) := 2\beta + 2\gamma - 1 - \beta x / \alpha.$$

Some Results for a Globally Lipschitz Drift Term

We first prove some results assuming that the drift term, σ is globally Lipschitz:

$$|\sigma(x) - \sigma(y)| \leq L(\sigma)|x - y|, \quad x, y \in \mathbb{R}^d.$$

We also define

Proposition

Suppose that $d < 2\alpha$ and $\rho = \rho(d) > 0$. Then there exists a universal constant $K := K(\alpha, \beta, \gamma, d) > 0$ such that for any

$$a > (8L^2(\sigma)K^2)^{1/\rho} \quad \text{and} \quad p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2}\right],$$

Then we have that

$$N_{a,p}(u) \leq 2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \quad (0.3)$$

where

$$\mathcal{T}_0 = \begin{cases} (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|u_0\|_\infty & \beta \in (0, 1] \\ (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases}$$

Moreover for any $T > 0$,

$$\sup_{x \in \mathbb{R}^d} \mathbb{E}(|u(t, x)|^p) \leq \exp(apt) \left[2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \right]^p, \quad t \in [0, T]. \quad (0.4)$$

Suppose that $0 < \theta < (\Theta - d) \wedge 2$ and $\rho = \rho(d) > 0$. Let $T > 0$ and $t, r \in [0, T]$. Then for any

$$a > (8L^2(\sigma)K^2)^{1/\rho} \quad \text{and} \quad p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2}\right],$$

and any $x, z \in \mathbb{R}^d$ we have that

$$\frac{\|u(t, x) - u(r, z)\|_p}{(|t - r|^q + |x - z|^\theta)^{1/2}} \leq C(p, \theta, T) \left[\mathcal{M}_1 + \mathcal{M}_2 e^{aT} \left(2 + \frac{c(\sigma)}{L(\sigma)} \right) \right]. \quad (0.5)$$

where,

1. $0 < q \leq \rho$ under $\beta \in (0, 1]$ and $\gamma \in (0, 1 - \beta]$,
2. $0 < q < \rho$ under $\beta \in (0, 2)$ and $\gamma > 0$,

and

$$\mathcal{M}_1 = \sqrt{p}c(\sigma) \quad \text{and} \quad \mathcal{M}_2 = \sqrt{p}L(\sigma).$$

Holder Continuity

The solution, u , has a version, still denoted by u , that is η_1 -Hölder continuous in time and η_2 -Hölder continuous in space with $0 < \eta_1 < q/2$ and $0 < \eta_2 < \theta/2$ where $0 < \theta < (\Theta - d) \wedge 2$ and

1. $0 < q \leq \rho$ under $\beta \in (0, 1]$ and $\gamma \in (0, 1 - \beta]$,
2. $0 < q < \rho$ under $\beta \in (0, 2)$ and $\gamma > 0$,

Moreover, if we consider

$$a > (8L^2(\sigma)K^2)^{1/\rho} \quad \text{and} \quad p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2}\right],$$

then we have that

$$\begin{aligned} & \mathbb{E} \left[\sup_{t \in [0, T], |x| \leq R} |u(t, x)|^p \right] \\ & \leq 2^{p-1} + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2^p e^{apT} \left(2 + \frac{c(\sigma)}{L(\sigma)} \right)^p \right]. \end{aligned}$$

Locally Lipschitz Diffusion Term

$$\begin{cases} (\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r [\sigma(u(t, x)) \dot{W}(x)] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b \in (0, 1] \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b \in (1, 2) \end{cases}$$

$$|\sigma(x)| \leq \sigma_1 + \sigma_2 |x| [\ln(|x|)]^\delta,$$

where $\sigma_1 = \sigma(0)$ and $\sigma_2 > 0$.

Theorem

Let $M, T > 0$. Under Dalang's condition, $\rho(d) > 0$ and $d < 2\alpha$, there exists a random field solution for $|x| \leq M$ denoted as $(u(t, x) : (t, x) \in [0, T] \times [-M, M])$. This solution is unique and satisfies

$$\sup_{t \in [0, T], x \in [-R, R]} |u(t, x)| < \infty, \text{ a.s.}$$

Proof Sketch

We consider a truncated version of the equation where σ replaced by σ_N :

$$\sigma_N(x) = \sigma(x)\mathbf{1}_{\{|x| \leq N\}} + \sigma(N)\mathbf{1}_{\{x > N\}} + \sigma(-N)\mathbf{1}_{\{x \leq -N\}}.$$

We see that σ_N is Lipschitz:

$$|\sigma_N(x)| \leq \sigma_1 + \sigma_2 \ln(2N)^\delta |x|,$$

so there exists a unique solution which is Hölder continuous, which we denote as $u_N := \{u_N(t, x) : (t, x) \in [0, T] \times \mathbb{R}\}$. We define,

$$\tau_N := \inf \left\{ t > 0 : \sup_{|x| \leq R} |u_N(t, x)| \geq N \right\} \wedge T.$$

We see that on $\{t < \tau_N\}$, $u_N(t, x) = u_{N+k}(t, x)$ for any $k \in \mathbb{N}$.

Proof Sketch

We can show that $\sup_N \tau_N = T$ which implies that $\{t < \tau_N\} \uparrow \Omega$. On each $\{t < \tau_N\}$, define $(u(t, x) : (t, x) \in [0, T) \times \mathbb{R})$ by $u(t, x) = u_N(t, x)$ and hence $u(t, x) = u_{N+k}(t, x)$ for any $k \in \mathbb{N}$. This implies that on $\{t < \tau_N\}$

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy \, Y(t-s, x-y) \sigma_N(u(s, y)) W(ds, dy).$$

However, on $\{t < \tau_N\}$, we have that $\sigma_N(u_N(t, x)) = \sigma(u(t, x))$ and so u satisfies

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy \, Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy), \quad \{t < \tau_N\}.$$

Lastly, since $\{t < \tau_N\} \uparrow \Omega$, we conclude that

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy \, Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy), \quad (t, x) \in [0, T) \times \mathbb{R} \quad (0.6)$$

References

 L. Chen, Y. Hu, D. Nualart (2019)

Nonlinear stochastic time-fractional slow and fast diffusion equations on $\mathbb{R}^d$.
Stochastic Processes and their Applications, 129 5073–5112.